

A Novel and Real-Time 6-dof Pose Estimation Method Using a 2-D Laser Profiler in a Single Scan

Tao Chen , Member, IEEE, Jia-Xin Liu, Yao-Yang Tsai, and Pei-Chun Lin , Senior Member, IEEE

Abstract—This work proposes a novel real-time six-degree-of-freedom (dof) pose estimation method using single-scan data from a 2-D laser profiler (LPF), providing a cost-effective alternative to traditional vision sensors in robotics. By scanning a custom-designed four-plane gauge, the 3-dof orientation and position relative to the gauge are sequentially computed using a low computational time solution, enabling high accuracy and low latency pose estimation on a standard computer. This method bridges the gap between low-accuracy camera systems and high-cost laser tracker systems, expanding LPF applications in real-time tasks. Sensitivity analysis was performed to guide the geometric design of the gauge to enhance robustness against fabrication errors. Simulations of structured and unstructured noises, including measurement noise, occlusions, and reflections, demonstrated the reliability of the method. Real-world experiments in cluttered environments further validated its robustness. Two fabricated gauges with different geometries confirmed the design principles, with a motion capture system as the ground truth. A hand-eye calibration experiment demonstrated the applicability of the method in robotic systems. Comparative benchmarks against a camera-based method revealed superior pose accuracy, lower computational cost, and improved calibration stability. These results establish LPFs as reliable, high-performance sensors for real-time 6-dof pose estimation in robotic vision applications.

Index Terms—6-degree of freedom (dof), laser profiler (LPF), manipulators, pose estimation.

I. INTRODUCTION

SIX-DEGREE-OF-FREEDOM (dof) pose estimation, which determines the position and orientation of a sensor relative to a known reference frame, is fundamental to fields, such as

Received 16 June 2025; revised 1 November 2025; accepted 21 February 2026. Recommended by Technical Editor W. Wang and Senior Editor J. Ueda. This work was supported by the National Science and Technology Council (NSTC), Taiwan under Grant 109-2634-F-002-039- and Grant 110-2221-E-002-038-. (Corresponding author: Pei-Chun Lin.)

The authors are with the Department of Mechanical Engineering, National Taiwan University, Taipei 10617, Taiwan (e-mail: peichunlin@ntu.edu.tw).

This article has supplementary material provided by the authors and color versions of one or more figures available at <https://doi.org/10.1109/TMECH.2026.3678182>.

Digital Object Identifier 10.1109/TMECH.2026.3678182

1083-4435 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See <https://www.ieee.org/publications/rights/index.html> for more information.

TABLE I

COMPARISON OF SENSORS UTILIZED IN THE 6-DOF POSE ESTIMATIONS OF THE MANIPULATOR

Sensors	Characteristics			
	Accuracy	Cost	Portability	Eye-in-Hand
LTS	0.5 $\mu\text{m}/\text{m}$	High	Medium	No
O-CMM	30 μm	Medium	Medium	No
Camera	150 μm	Low	High	Yes
C-CMM	0.5 μm	High	Low	No
LPF	5 μm	Medium	High	Yes

Note: Laser tracker system (LTS), optical coordinate measuring machine (O-CMM), cameras, contact coordinate measuring machine (C-CMM), and 2-D LPF. The table is partially adapted from [23] and [24].

autonomous navigation [1], simultaneous localization and mapping [2], [3], augmented and virtual reality [4], and robotics [5]. A diverse range of sensors, including GPS, Wi-Fi, lidar, and ultrasound, have been employed for this task [6], [7]. Among these sensors, cameras remain the most affordable and versatile, driving their widespread adoption in both research and industry [8], [9]. Notable advancements include methods for large-scale environment mapping [10], [11] and adaptation to dynamic scenes [12]. Beyond standard 2-D imaging, specialized systems, such as RGB-D cameras [13], stereo-camera systems [14], and event-based cameras [15], have further enhanced estimation capabilities.

In robotics, 6-dof pose estimation is critical for applications, such as mobile robot navigation [16] and tool center point tracking in surgical [17] and construction robotics [18]. However, high-precision tasks, such as machining, assembly, and real-time trajectory compensation, demand greater accuracy than conventional camera systems can deliver. To meet these demands, various 6-dof sensing technologies have been adopted, as summarized in Table I, which compares their accuracy, cost, and portability. High-accuracy systems, including laser tracker systems (LTSs), contact coordinate measuring machines (C-CMMs), and optical CMMs (O-CMMs), typically operate in an “eye-to-hand” configuration, requiring the tool center point to remain within a fixed field of view. While LTSs provide state-of-the-art accuracy [19], [20], their high cost is often prohibitive. Conversely, O-CMMs provide a more cost-effective solution for tasks with moderate accuracy requirements [21], [22]. “Eye-in-hand” systems, such as cameras and 2-D laser profilers (LPFs), are mounted directly on the robot. Although cameras are

widely used for precalibration and object detection, their utility for high-frequency, high-accuracy trajectory compensation is limited by resolution, frame rate, and computational overhead, despite their inherent 6-dof estimation capabilities [23]. LPFs, serving as a middle ground between these two sensor types, provide higher accuracy and sampling rates than cameras at a lower cost than LTSs. However, despite its importance in other optical sensing domains [23], [25], [26], [27], real-time 6-dof pose estimation using a single LPF scan has not been previously addressed.

One critical application for real-time 6-dof pose estimation in robotics is hand-eye calibration, which computes the rigid transformation between a robot's tool center point and a mounted sensor. Conventional camera-based calibration algorithms [28], [29] are incompatible with LPFs, which output a 2-D cross-sectional profile ($1 \times N$ height data and $1 \times N$ horizontal positions). This has motivated the development of specialized LPF calibration methods, including those based on multisensor fusion [30], [31], planar constraints [32], [33], [34], and straight-edge features [35]. While effective, these techniques often require multiple scans, precluding real-time operation. This study addresses this gap by developing a real-time 6-dof pose estimation method tailored to LPF data, enabling their seamless integration into standard calibration pipelines and expanding their utility in time-sensitive robotic tasks.

The core innovation of this research is a custom-designed four-plane gauge that imposes geometric constraints sufficient for 6-dof pose recovery from a single LPF scan. We developed a theoretical model for this process and optimized the gauge geometry via sensitivity analysis to minimize the impact of fabrication errors. The robustness of the method was evaluated through simulations involving measurement noise, occlusions, and specular reflections. Experimental validation was performed using a Vicon motion capture system to confirm the sensitivity analysis and evaluate the performance of hand-eye calibration on a six-axis manipulator. Finally, a benchmark comparison against a camera-based method demonstrates the proposed method to be a high-performance, cost-effective approach for real-time robotic vision.

The rest of this article is organized as follows. Section II defines the problem statement. Section III details the theoretical model and its solution. Section IV presents the gauge design and sensitivity analysis, and robustness evaluation. Section V describes the experimental validation and hand-eye calibration application. Section VI compares the proposed method with a camera-based approach. Finally, Section VII concludes this article.

II. PROBLEM STATEMENT

In this study, we developed a real-time method to estimate the 6-dof pose of a 2-D LPF relative to a custom gauge from a single scan. The primary objective was to compute the homogeneous transformation between the sensor frame of the LPF and reference frame of the gauge using single-scan data from the LPF mounted on the flange of a robotic manipulator. As illustrated

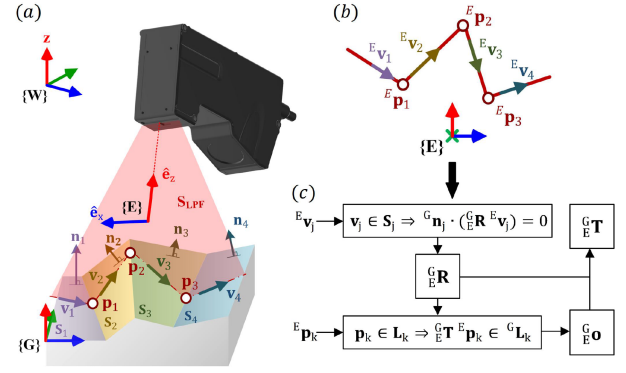


Fig. 1. (a) Definitions of the coordinate frames and related transformations. (b) Captured profile showing identified direction vectors and corner points. (c) Workflow for computing the LPF pose.

in Fig. 1(a), the problem involves three key coordinate frames: the world frame $\{\mathbf{W}\}$, LPF sensor frame $\{\mathbf{E}\}$, and gauge frame $\{\mathbf{G}\}$.

The LPF frame, $\{\mathbf{E}\}$, is defined by the measurement axes of the sensor: Its x -axis, $\hat{\mathbf{e}}_x$, aligns with the laser scan line, and its z -axis, $\hat{\mathbf{e}}_z$, corresponds to the depth measurement direction. All profile data points are captured in this frame. The gauge frame, $\{\mathbf{G}\}$, is fixed to the custom gauge, with its origin at a designated corner and its axes aligned with the orthogonal edges of the gauge, as shown in Fig. 1(a). With these definitions, the poses of the LPF and gauge relative to the world frame $\{\mathbf{W}\}$ are described according to their respective homogeneous transformations

$${}^W_E \mathbf{T} = \begin{bmatrix} {}^W_E \mathbf{R} & {}^W_E \mathbf{o} \\ \mathbf{0}^\top & 1 \end{bmatrix} \quad (1)$$

$${}^W_G \mathbf{T} = \begin{bmatrix} {}^W_G \mathbf{R} & {}^W_G \mathbf{o} \\ \mathbf{0}^\top & 1 \end{bmatrix} \quad (2)$$

where $\mathbf{R} \in SO(3)$ and $\mathbf{o} \in \mathbb{R}^3$ represent the rotation matrix and origin, respectively, of a given frame relative to $\{\mathbf{W}\}$. The vector $\mathbf{0} \in \mathbb{R}^3$ is the zero vector. The homogeneous transformation from the LPF frame $\{\mathbf{E}\}$ to the gauge frame $\{\mathbf{G}\}$, denoted by ${}^G_E \mathbf{T}$, is then derived as

$${}^G_E \mathbf{T} = \begin{bmatrix} {}^G_E \mathbf{R} & {}^G_E \mathbf{o} \\ \mathbf{0}^\top & 1 \end{bmatrix} = ({}^W_G \mathbf{T})^{-1} {}^W_E \mathbf{T}. \quad (3)$$

The transformation ${}^G_E \mathbf{T}$ represents the 6-dof pose of the LPF relative to the gauge. The primary objective of this work was to estimate this transformation in real time from a single LPF scan. To achieve this, a two-step approach was adopted. First, the orientation ${}^G_E \mathbf{R}$ was estimated, and second, the position ${}^G_E \mathbf{o}$ was computed using the previously determined orientation.

III. THEORETICAL MODEL

This section details the sequential method for determining the 6-dof pose of the LPF, ${}^G_E \mathbf{T}$, from a single scan of a custom-designed four-plane gauge. The estimation process, as illustrated

in Fig. 1(c), first computes the orientation, ${}^G_E\mathbf{R}$, and then the origin, ${}^G_E\mathbf{o}$, as defined in (3).

To facilitate the derivation, the following notation is defined, with physical interpretations illustrated in Fig. 1(a). The four planes of the gauge are denoted by $\{\mathbf{S}_j\}_{j=1,2,3,4}$, with corresponding unit normal vectors $\{\mathbf{n}_j\}_{j=1,2,3,4}$. The LPF captures a 2-D profile that lies on a single plane (i.e., the profile plane) $\mathbf{S}_{\text{LPF}}$, which is coincident with the xz -plane of the LPF frame $\{\mathbf{E}\}$ (i.e., all y -coordinates are zero), and has a normal vector $\mathbf{n}_{\text{LPF}}$. The scanned profile consists of four line segments, whose unit direction vectors are $\{\mathbf{v}_j\}_{j=1,2,3,4}$, and three corner points, $\{\mathbf{p}_k\}_{k=1,2,3}$, located at the intersections of the adjacent segments. These features, expressed in the LPF frame $\{\mathbf{E}\}$, are denoted as $\{{}^E\mathbf{v}_j\}$ and $\{{}^E\mathbf{p}_k\}$, as shown in Fig. 1(b).

The corner points are initially extracted from the raw profile data using the Ramer–Douglas–Peucker (RDP) algorithm [36] for its computational efficiency. The unit direction vectors are then obtained by applying linear regression to the points within each segment. Finally, the corner point positions are refined by computing the intersections of the adjacent regression lines, a step that minimizes the effect of measurement noise at the physical edges of the gauge. These extracted direction vectors and corner points are geometrically constrained by the gauge design, enabling the pose solution outlined in Fig. 1(c). Further conceptual details of the method are available in Section II-B of the Supplementary Material.

A. Orientation Model

This section presents the derivation of the LPF frame's orientation, ${}^G_E\mathbf{R}$. Each direction vector of a scanned line segment, $\mathbf{v}_j$, as depicted in Fig. 1(a), has representations in both the LPF frame (${}^E\mathbf{v}_j$) and gauge frame (${}^G\mathbf{v}_j$). These representations are related by the unknown rotation matrix

$${}^G\mathbf{v}_j = {}^G_E\mathbf{R} {}^E\mathbf{v}_j. \quad (4)$$

The vectors ${}^E\mathbf{v}_j$ are extracted directly from the profile data, which are captured in the LPF frame $\{\mathbf{E}\}$, as depicted in Fig. 1(b). The vectors ${}^G\mathbf{v}_j$ and orientation ${}^G_E\mathbf{R}$ are the unknown variables to be determined.

As shown in Fig. 1(a), each vector $\mathbf{v}_j$ is formed by the intersection of the LPF profile plane, $\mathbf{S}_{\text{LPF}}$, and a corresponding gauge plane, $\mathbf{S}_j$. Consequently, $\mathbf{v}_j$ must be orthogonal to the normal vector $\mathbf{n}_j$ of its respective gauge plane. This geometric constraint, expressed in the gauge frame, is formulated by taking the inner product of both sides of (4) with ${}^G\mathbf{n}_j$

$${}^G\mathbf{n}_j \cdot {}^G\mathbf{v}_j = {}^G\mathbf{n}_j \cdot ({}^G_E\mathbf{R} {}^E\mathbf{v}_j) = 0, \quad j = 1, 2, 3, 4. \quad (5)$$

The orientation of the LPF frame ${}^G_E\mathbf{R}$, is expressed using a ZXY Tait–Bryan angle sequence. In this sequence, the y -axis rotation is applied last, implying that after the initial z - and x -axes rotations, the transformed profile lies in a plane that is coplanar with the actual scanned profile but deviates from it by an angle θ_y . The final rotation, $\mathbf{R}_y(\theta_y)$, is then applied to align each segment vector ${}^E\mathbf{v}_j$ of the captured profile with its corresponding gauge plane $\mathbf{S}_j$. In other words, the full rotation ensures that each transformed vector lies within its respective

gauge plane. On the basis of this formulation, the orientation matrix is decomposed into the product $\mathbf{R}_z(\theta_z)\mathbf{R}_x(\theta_x)\mathbf{R}_y(\theta_y)$, and the rotation angles are solved sequentially, starting with θ_y , followed by a simultaneous estimation of θ_z and θ_x .

As $\mathbf{R}_y(\theta_y)$ is the final rotation applied and eliminates the uniform angular deviation between the profile and the gauge planes, a single constraint suffices to solve for this dof. In the following derivation, the constraint of $j = 1$ was used to compute θ_y . Subsequently, the constraints of $j = 2, 3, 4$ were applied to solve for θ_x and θ_z . The solution for θ_y begins with rewriting (5) at $j = 1$ and expanding the orientation matrix ${}^G_E\mathbf{R}$ according to the ZXY Tait–Bryan angle convention, which gives

$${}^G\mathbf{n}_1 \cdot (\mathbf{R}_z(\theta_z)\mathbf{R}_x(\theta_x)\mathbf{R}_y(\theta_y){}^E\mathbf{v}_1) = 0. \quad (6)$$

Consider the result of multiplying the z -axis and x -axis rotation matrices

$$\mathbf{R}_z(\theta_z)\mathbf{R}_x(\theta_x) = \begin{bmatrix} c\theta_z & -c\theta_x s\theta_z & s\theta_x s\theta_z \\ s\theta_z & c\theta_x c\theta_z & -s\theta_x c\theta_z \\ 0 & s\theta_x & c\theta_x \end{bmatrix} \quad (7)$$

where c and s denote the cosine and sine functions, respectively. The third element of the first column is zero, indicating that if the z -component of the resulting vector of $\mathbf{R}_y(\theta_y){}^E\mathbf{v}_1$ is zero, then it remains unaffected after applying the $\mathbf{R}_z(\theta_z)\mathbf{R}_x(\theta_x)$ rotation, as the y -component is always zero. To exploit this property, the gauge frame is defined such that its xy -plane coincides with the gauge plane $\mathbf{S}_1$, resulting in the normal vector ${}^G\mathbf{n}_1 = [0 \ 0 \ 1]^\top$. Under this definition, the constraint expressed in (6) is automatically satisfied for arbitrary $\mathbf{R}_z(\theta_z)\mathbf{R}_x(\theta_x)$ as long as $\mathbf{R}_y(\theta_y){}^E\mathbf{v}_1$ has a zero z -component. Therefore, (6) simplifies to a constraint involving only the z -component of the rotated vector $\mathbf{R}_y(\theta_y){}^E\mathbf{v}_1$, written as

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^\top \cdot (\mathbf{R}_y(\theta_y){}^E\mathbf{v}_1) = 0. \quad (8)$$

This means θ_y is the angle that rotates ${}^E\mathbf{v}_1$ onto the gauge plane $\mathbf{S}_1$, which corresponds to the xy -plane of the gauge frame. This ensures that the z -component of the rotated vector, $\mathbf{R}_y(\theta_y){}^E\mathbf{v}_1$, is zero, satisfying the constraint in (8). Specifically, θ_y is computed as a signed angle between the vector ${}^E\mathbf{v}_1$ and the plane spanned by the basis vectors $\hat{\mathbf{g}}_x = [1 \ 0 \ 0]^\top$ and $\hat{\mathbf{g}}_y = [0 \ 1 \ 0]^\top$, which represent the x - and y -axes of the gauge frame, respectively. The solution for θ_y is then given as follows:

$$\theta_y = \text{sign}({}^E\mathbf{v}_1 \times \hat{\mathbf{g}}_y) \cdot \hat{\mathbf{g}}_x (\arccos({}^E\mathbf{v}_1 \cdot \hat{\mathbf{g}}_x)). \quad (9)$$

With θ_y known, the problem is reduced to a 2-dof estimation for the rotation $\mathbf{R}_z(\theta_z)\mathbf{R}_x(\theta_x)$. While an analytical solution exists, which is derived by finding a common root of two nonlinear equations, practical issues, such as sensor noise and fabrication errors, may prevent it from yielding valid results. Therefore, a numerical optimization approach is adopted using least-squares optimization to find the θ_z and θ_x that best satisfy the remaining three constraints ($j = 2, 3, 4$). This approach is well suited for noisy, overdetermined systems and is computationally efficient for real-time applications. Details of the analytical solution are provided in Section II-A of the Supplementary Material.

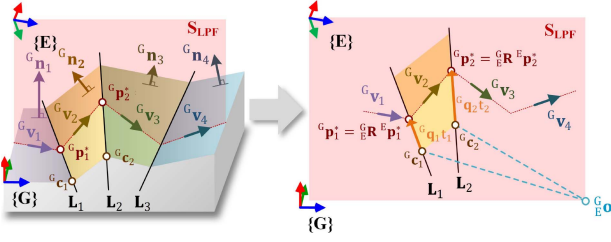


Fig. 2. Origin of the LPF frame ${}^G_E\mathbf{o}$ is solved by at least two linear constraints formed by the three planes of the gauge.

The numerical approach starts with the constraint equations derived from the three planes— S_2 , S_3 , and S_4 —expressed as

$${}^G\mathbf{n}_j \cdot (\mathbf{R}_z(\theta_z)\mathbf{R}_x(\theta_x)\mathbf{R}_y(\theta_y){}^E\mathbf{v}_j) = 0 \quad (10)$$

for $j = 2, 3, 4$ and is further simplified by defining the segment vector after the y -axis rotation [with θ_y already solved in (9)] as

$${}^E\mathbf{v}'_j = \mathbf{R}_y(\theta_y){}^E\mathbf{v}_j. \quad (11)$$

Substituting this into (10) allows us to merge the known terms and rewrite the equation in the simplified form

$${}^G\mathbf{n}_j \cdot (\mathbf{R}_z(\theta_z)\mathbf{R}_x(\theta_x){}^E\mathbf{v}'_j) = 0. \quad (12)$$

Accordingly, the problem is formulated as a nonlinear least-squares optimization that estimates the optimal values of θ_z and θ_x by minimizing the residuals of (12). The objective function is given by the following equation:

$$\operatorname{argmin}_{\theta_z, \theta_x} \sum_{j=2}^4 \|{}^G\mathbf{n}_j \cdot ((\mathbf{R}_z(\theta_z)\mathbf{R}_x(\theta_x)){}^E\mathbf{v}'_j)\|_2^2. \quad (13)$$

Once θ_z and θ_x are obtained, the full orientation of the LPF frame, ${}^G_E\mathbf{R}$, can be constructed as follows:

$${}^G_E\mathbf{R} = \mathbf{R}_z(\theta_z)\mathbf{R}_x(\theta_x)\mathbf{R}_y(\theta_y). \quad (14)$$

B. Position Model

To solve the position vector ${}^G_E\mathbf{o}$, as shown in (3), the intersecting lines between the adjacent gauge planes, S_j and S_{j+1} for $j = 1, 2, 3$ are utilized. As illustrated in Fig. 2, the black lines that lie on the gauge and between the planes S_j and S_{j+1} are denoted as $L_{k=j}$. The orange arrows in Fig. 2 are the unit direction vectors of each line, ${}^G\mathbf{q}_k$, expressed in the gauge frame. They are obtained as the normalized cross product of the corresponding plane normals: ${}^G\mathbf{n}_k \times {}^G\mathbf{n}_{k+1}$. A point on the line, ${}^G\mathbf{c}_k$, illustrated as the brown points in Fig. 2, is determined from the known geometry of the gauge. Hence, each line expressed in the gauge frame can be parameterized as

$$\begin{aligned} {}^G\mathbf{L}_k : {}^G\mathbf{p}_k(t_k) &= \frac{({}^G\mathbf{n}_k \times {}^G\mathbf{n}_{k+1})}{\|({}^G\mathbf{n}_k \times {}^G\mathbf{n}_{k+1})\|} t_k + {}^G\mathbf{c}_k \\ &= {}^G\mathbf{q}_k t_k + {}^G\mathbf{c}_k \end{aligned} \quad (15)$$

for $k = 1, 2, 3$, where $t_k \in \mathbb{R}$ is a scalar parameter. Thus, the three intersecting lines formed by the four planes are expressed

as follows:

$$\begin{cases} {}^G\mathbf{L}_1 : {}^G\mathbf{q}_1 t_1 + {}^G\mathbf{c}_1 \\ {}^G\mathbf{L}_2 : {}^G\mathbf{q}_2 t_2 + {}^G\mathbf{c}_2 \\ {}^G\mathbf{L}_3 : {}^G\mathbf{q}_3 t_3 + {}^G\mathbf{c}_3. \end{cases} \quad (16a)$$

$$\begin{cases} {}^G\mathbf{L}_1 : {}^G\mathbf{q}_1 t_1 + {}^G\mathbf{c}_1 \\ {}^G\mathbf{L}_2 : {}^G\mathbf{q}_2 t_2 + {}^G\mathbf{c}_2 \\ {}^G\mathbf{L}_3 : {}^G\mathbf{q}_3 t_3 + {}^G\mathbf{c}_3. \end{cases} \quad (16b)$$

$$\begin{cases} {}^G\mathbf{L}_1 : {}^G\mathbf{q}_1 t_1 + {}^G\mathbf{c}_1 \\ {}^G\mathbf{L}_2 : {}^G\mathbf{q}_2 t_2 + {}^G\mathbf{c}_2 \\ {}^G\mathbf{L}_3 : {}^G\mathbf{q}_3 t_3 + {}^G\mathbf{c}_3. \end{cases} \quad (16c)$$

Each line L_k intersects the profile plane S_{LPF} at a specific point ${}^G\mathbf{p}_k^*$ in the gauge frame, which corresponds to the corner point ${}^E\mathbf{p}_k$ in the captured profile data. These two representations of the same physical point—expressed in the gauge and LPF frames, respectively—are related by the LPF pose, that is, the transformation from the LPF frame to the gauge frame. This transformation is defined by the rotation matrix ${}^G_E\mathbf{R}$ and the translation vector ${}^G_E\mathbf{o}$

$${}^G\mathbf{p}_k^* = {}^G_E\mathbf{T}^E\mathbf{p}_k = {}^G_E\mathbf{R}^E\mathbf{p}_k + {}^G_E\mathbf{o}. \quad (17)$$

The point ${}^E\mathbf{p}_k$ is known a priori from the captured profile, and the orientation ${}^G_E\mathbf{R}$ was determined based on the derivation in Section III-A. Substituting ${}^G\mathbf{p}_k^*$ with (15) and rearranging the terms yield the following:

$${}^G\mathbf{q}_k t_k^* + ({}^G\mathbf{c}_k - {}^G_E\mathbf{R}^E\mathbf{p}_k) = {}^G_E\mathbf{o} \quad (18)$$

where t_k^* is the scalar parameter corresponding to the point ${}^G\mathbf{p}_k^*$. Applying (18) for $k = 1, 2, 3$ resulted in the following three equations, each corresponding to one intersection line, forming a system that constrains the translation vector ${}^G_E\mathbf{o}$:

$$\begin{cases} {}^G\mathbf{q}_1 t_1^* + ({}^G\mathbf{c}_1 - {}^G_E\mathbf{R}^E\mathbf{p}_1) = {}^G_E\mathbf{o} \\ {}^G\mathbf{q}_2 t_2^* + ({}^G\mathbf{c}_2 - {}^G_E\mathbf{R}^E\mathbf{p}_2) = {}^G_E\mathbf{o} \\ {}^G\mathbf{q}_3 t_3^* + ({}^G\mathbf{c}_3 - {}^G_E\mathbf{R}^E\mathbf{p}_3) = {}^G_E\mathbf{o}. \end{cases} \quad (19a)$$

$$\begin{cases} {}^G\mathbf{q}_1 t_1^* + ({}^G\mathbf{c}_1 - {}^G_E\mathbf{R}^E\mathbf{p}_1) = {}^G_E\mathbf{o} \\ {}^G\mathbf{q}_2 t_2^* + ({}^G\mathbf{c}_2 - {}^G_E\mathbf{R}^E\mathbf{p}_2) = {}^G_E\mathbf{o} \\ {}^G\mathbf{q}_3 t_3^* + ({}^G\mathbf{c}_3 - {}^G_E\mathbf{R}^E\mathbf{p}_3) = {}^G_E\mathbf{o}. \end{cases} \quad (19b)$$

$$\begin{cases} {}^G\mathbf{q}_1 t_1^* + ({}^G\mathbf{c}_1 - {}^G_E\mathbf{R}^E\mathbf{p}_1) = {}^G_E\mathbf{o} \\ {}^G\mathbf{q}_2 t_2^* + ({}^G\mathbf{c}_2 - {}^G_E\mathbf{R}^E\mathbf{p}_2) = {}^G_E\mathbf{o} \\ {}^G\mathbf{q}_3 t_3^* + ({}^G\mathbf{c}_3 - {}^G_E\mathbf{R}^E\mathbf{p}_3) = {}^G_E\mathbf{o}. \end{cases} \quad (19c)$$

This system can be rewritten in standard form as $\mathbf{A}\mathbf{x} = \mathbf{b}$, with $\mathbf{A} \in \mathbb{R}^{3 \times 3}$ and $\mathbf{x} \in \mathbb{R}^{3 \times 1}$

$$\begin{aligned} \begin{bmatrix} 2{}^G\mathbf{q}_1 & -{}^G\mathbf{q}_2 & -{}^G\mathbf{q}_3 \end{bmatrix} \begin{bmatrix} t_1^* \\ t_2^* \\ t_3^* \end{bmatrix} \\ = {}^G_E\mathbf{R}(2{}^E\mathbf{p}_1 - {}^E\mathbf{p}_2 - {}^E\mathbf{p}_3) - (2{}^G\mathbf{c}_1 - {}^G\mathbf{c}_2 - {}^G\mathbf{c}_3) \end{aligned} \quad (20)$$

where $\begin{bmatrix} t_1^* & t_2^* & t_3^* \end{bmatrix}^\top$ is the unknown vector $\mathbf{x}$. Using the pseudoinverse, the solution $\mathbf{x}^*$ is given by

$$\mathbf{x}^* = \begin{bmatrix} t_1^* & t_2^* & t_3^* \end{bmatrix}^\top = \mathbf{A}^\dagger \mathbf{b}. \quad (21)$$

Substituting the computed values of t_k^* into (19) yields three estimates of the LPF origin, denoted ${}^G_E\mathbf{o}_1$, ${}^G_E\mathbf{o}_2$, and ${}^G_E\mathbf{o}_3$. To mitigate the estimation errors, the final origin ${}^G_E\mathbf{o}$ is computed as their average

$${}^G_E\mathbf{o} = \frac{({}^G_E\mathbf{o}_1 + {}^G_E\mathbf{o}_2 + {}^G_E\mathbf{o}_3)}{3}. \quad (22)$$

Combining the resulting ${}^G_E\mathbf{o}$ with the orientation ${}^G_E\mathbf{R}$ derived in (14), the 6-dof pose of the LPF frame relative to the gauge frame ${}^G_E\mathbf{T}$, as shown in (3), is solved.

IV. ROBUSTNESS AND NOISE ANALYSIS

Real-world deployment of the proposed 6-dof pose estimation system must account for inevitable noise and disturbances that

can degrade performance. These factors include 1) unstructured noise, such as random measurement fluctuations and gauge fabrication errors, 2) structured noise, such as occlusions or specular reflections, which cause systematic profile distortions, and 3) environmental disturbances, such as obstacles in cluttered workspaces, which may be captured in the scan. This section evaluates the robustness of the method against these conditions through a combination of simulation and experimental validation.

A. Geometry Design and Sensitivity Analysis of the Gauge

The unstructured noise in the system primarily originates from the measurement noise of the LPF and the fabrication error of the gauge. As the gauge is custom-designed, its geometry can be optimized to reduce sensitivity to both fabrication error and measurement noise, thereby enhancing the overall accuracy and robustness of the pose estimation system. To investigate the influence of geometry on sensitivity, a large-scale simulation-based analysis was performed, beginning with the formulation of modeling assumptions, followed by evaluation across a range of randomly generated geometries. The analysis is simplified by our frame definition. As the gauge frame $\{G\}$ is defined with its origin on plane S_1 and its z -axis aligned with the plane's normal, ${}^G\mathbf{n}_1$ is constant (${}^G\mathbf{n}_1 = [0 \ 0 \ 1]^T$). Thus, the design problem is reduced to determining the optimal configuration of the three remaining normal vectors: ${}^G\mathbf{n}_2$, ${}^G\mathbf{n}_3$, and ${}^G\mathbf{n}_4$. Further details on the modeling are available in Section III-A of the Supplementary Material.

The analysis considered a broad range of gauge geometries. In the simulation, 10 000 unique gauge geometries were generated by randomly sampling sets of plane normals, ${}^G\{\mathbf{n}_j\}_{j=1,2,3,4}$, representing a wide variety of spatial arrangements. For each configuration, 50 independent sets of random fabrication errors δ were applied. These errors were modeled as angular perturbations following a Gaussian distribution with a zero mean and a standard deviation of 1° . This resulted in 500 000 simulated 6-dof pose estimation cases. For each gauge geometry, the simulation computed the means and standard deviations of both orientation and position errors across 50 fabrication error instances. These performance indicators are denoted as $\Delta\theta_{\text{avg}}$, $\Delta\theta_{\text{std}}$, Δd_{avg} , and Δd_{std} , where $\Delta\theta$ represents the orientation error and Δd represents the distance error. To quantify the sensitivity of each gauge design, these errors were analyzed as functions of two key geometric parameters: ϕ and γ . The definitions and rationale for using these two geometric parameters in the analysis are explained as follows.

Considering that pose is estimated from the scanned profile, it is critical to identify profile shapes that are inherently less sensitive to fabrication errors. Intuitively, a symmetrical profile, such as one resembling a lateral triangle, may exhibit lower sensitivity. To quantify this relationship, we introduced a symmetry parameter, ϕ , which measures the symmetry of the gauge plane normals $\{{}^G\mathbf{n}_j\}_{j=2,3,4}$ with respect to the profile plane of the LPF. The normal of the LPF profile plane, when expressed in the gauge frame, is ${}^G\hat{\mathbf{e}}_y$. The parameter ϕ is defined as the

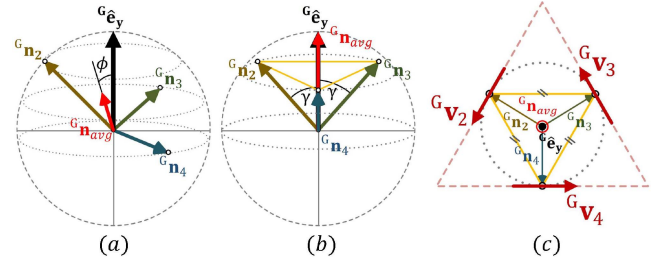


Fig. 3. (a) Parameter ϕ is defined as the angle between ${}^G\mathbf{n}_{\text{avg}}$ and ${}^G\hat{\mathbf{e}}_y$, and their spatial relations are shown in the figure. (b) Parameter γ is defined as the angle between ${}^G\hat{\mathbf{e}}_y$ and each of the normal vectors ${}^G\{\mathbf{n}_j\}_{j=2,3,4}$ when $\phi = 0^\circ$ or 180° . (c) Top view of (b), illustrating that the profile vectors form an equilateral triangle.

angle between this normal and the average of the three gauge plane normals, ${}^G\mathbf{n}_{\text{avg}} = \frac{1}{3} \sum_{j=2}^4 {}^G\mathbf{n}_j$, as illustrated in Fig. 3(a). A value of ϕ near 0° or 180° indicates that ${}^G\mathbf{n}_{\text{avg}}$ is nearly collinear with ${}^G\hat{\mathbf{e}}_y$. This corresponds to a configuration where the gauge planes are symmetrically distributed around the profile plane of the LPF, which we hypothesized to lead to a more robust estimation. The detailed mathematical formulation for ϕ is provided in Section III-B of the Supplementary Material.

For validating the suggested influence of ϕ on the sensitivity of the gauge, Gaussian noise with zero mean and a standard deviation of 1° was applied to the geometry of the gauge to simulate fabrication errors. Associated mathematical details can be found in Section III-C of the Supplementary Material.

The pose estimation error was computed by comparing the estimated pose with the fabrication error ${}^G\mathbf{T}_{\text{noisy}}$ against the ideal pose ${}^G\mathbf{T}_{\text{ideal}}$, under each of the 10 000 configurations of ${}^G\{\mathbf{n}_j\}_{j=2,3,4}$. For each configuration, 50 independent noise realizations were applied. Two error metrics, including the orientation and position errors, are used. The orientation error, $\Delta\theta$, quantifies the angular difference between the estimated and ground-truth rotation matrices

$$\Delta\theta = \arccos \left(\frac{\text{tr}({}^G\mathbf{R}_{\text{noisy}} {}^G\mathbf{R}_{\text{ideal}}^{-1}) - 1}{2} \right). \quad (23)$$

The position error, Δd , is the Euclidean distance between the estimated and ground-truth translation vectors

$$\Delta d = \| {}^G\mathbf{o}_{\text{noisy}} - {}^G\mathbf{o}_{\text{ideal}} \|_2. \quad (24)$$

The average and standard deviation of the resulting 50 sets of orientation and position errors were then computed and analyzed. The distribution of these error statistics across the 10 000 configurations is shown in Fig. 4(a).

The results proved the earlier suggestion by showing that both the orientation error $\Delta\theta$ and position error Δd exhibited lower maximum values and were more tightly concentrated near smaller values when ϕ was near 0° or 180° . These values correspond to configurations where the gauge planes are symmetrically distributed around the profile plane, as depicted in Fig. 3(b). This symmetry results in a scanned profile whose vectors ${}^G\{\mathbf{v}_j\}_{j=2,3,4}$ can form an equilateral triangle [illustrated by the red lines in Fig. 3(c)]. Thus, the results demonstrate that

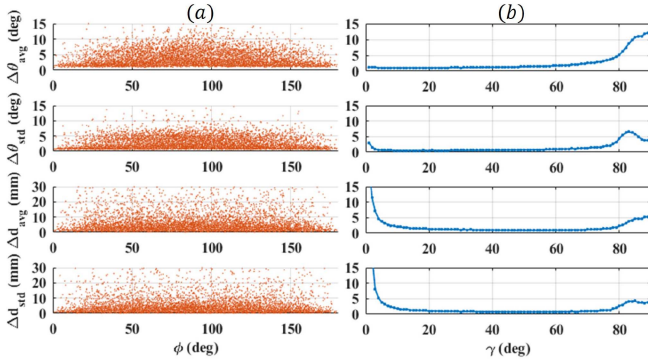


Fig. 4. Mean and standard deviation of the position and orientation errors that resulted from the fabrication errors of the gauge at (a) different ϕ values ranging from 0° to 180° and (b) different γ values ranging from 0° to 90° in simulation.

pose estimation is significantly more robust to fabrication errors when the normal vectors $^G\{\mathbf{n}_j\}_{j=2,3,4}$ exhibit higher symmetry with respect to the LPF profile plane.

With the optimal symmetry established ($\phi = 0^\circ$ or 180°), the influence of the plane divergence was investigated. A second geometric parameter, γ , is defined as the angle between the profile plane normal, $^G\hat{\mathbf{e}}_y$, and each of the gauge plane normals, $\{^G\mathbf{n}_j\}_{j=2,3,4}$, as shown in Fig. 3(b). Simulation evaluated how this parameter affects pose estimation accuracy by varying the γ values from 0° to 90° and observing the resulting changes in orientation error $\Delta\theta$ and Δd . The results, as shown in Fig. 4(b), indicate that both errors were minimized when $\gamma \approx 45^\circ$, indicating that this intermediate γ leads to a more robust configuration for the proposed 6-dof pose estimation. Sensitivity increased as the γ value approached the extreme values of 0° and 90° , suggesting higher sensitivity to fabrication errors at these extremes.

In summary, the simulation results indicate that the gauge exhibited lower sensitivity to fabrication error when $\phi = 0^\circ$ (or 180°), where the profile vectors $^G\{\mathbf{v}_j\}_{j=2,3,4}$ form an equilateral triangle, and when the parameter γ was approximately 45° . To validate these findings experimentally, two gauges were fabricated with parameters (ϕ, γ) set to $(0^\circ, 45^\circ)$ and $(0^\circ, 85^\circ)$, respectively. In addition, experimental evaluations of variations in the ϕ value were not conducted because ϕ does not correspond to a unique gauge geometry. For a given ϕ value, multiple gauge geometries can result in different sensitivity levels in both orientation and position errors, making it difficult to draw meaningful conclusions from direct comparisons across different ϕ values. Therefore, the experimental focus was placed on the symmetrical case ($\phi = 0^\circ$) to isolate the effect of varying ϕ values and directly test the influence of γ .

Finally, the physical design of the gauge must ensure that all four planes are simultaneously visible to the LPF in a single scan while avoiding sharp, fragile corners that are difficult to manufacture accurately. The final gauge designs reflecting these principles are shown in Fig. 6(a). Further details are provided in Section III-D of the Supplementary Material.

To evaluate the sensitivity of the system to the intrinsic measurement noise of the LPF, a second simulation was conducted.

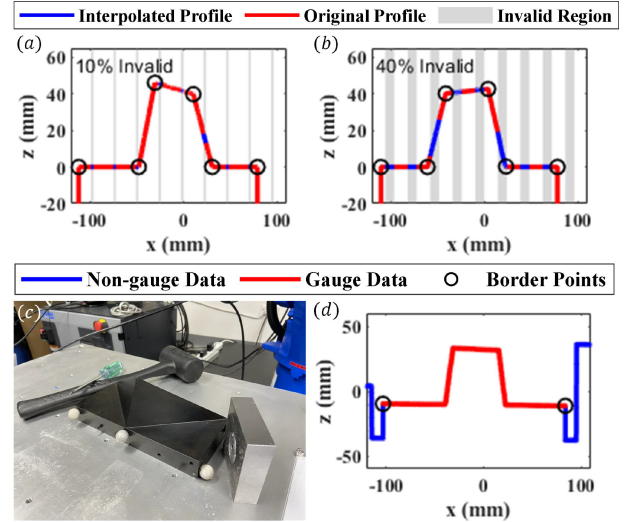


Fig. 5. Evaluation of the effects of the structured noise and cluttered workspace on the pose estimation qualities at (a) the well-interpolated profile and correct corner point detection at a low noise level (10%), (b) the poor-interpolated profile that leads to incorrect corner point detection at a high-noise level (40%), (c) the cluttered workspace with obstacles, and (d) the result of the corresponding captured profile applied with the gauge-identifying algorithm.

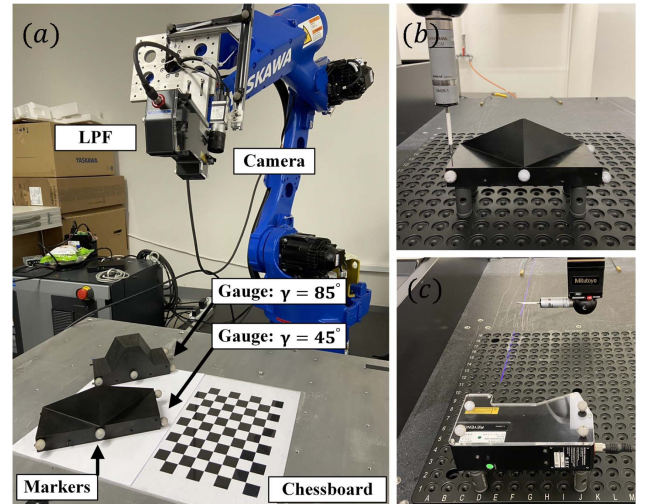


Fig. 6. (a) Experimental setup of the LPF-based measurement system (including the LPF, two gauges, and the retroreflective markers for VICON measurement) and camera-based system (including the camera and chessboard) used for benchmark. (b) and (c) Transformation matrix measurement between the gauge frame, LPF frame, and the frame constructed by markers using a CMM.

This analysis compared the performance of the optimal gauge design ($\{\phi, \gamma\} = \{0^\circ, 45^\circ\}$) with that of a suboptimal design ($\{\phi, \gamma\} = \{0^\circ, 85^\circ\}$). For each geometry, ideal profile data were generated for five random LPF poses. Gaussian noise with a standard deviation of $50 \mu\text{m}$, matching the specified repeatability of the LPF used in our experiments (see Section V-A), was then added to the profile data. For each pose, this process was repeated with ten independent noise realizations, yielding 50 noisy profiles per gauge design.

TABLE II

SIMULATION RESULTS OF THE ORIENTATION REPEATABILITY θ_{rp} AND POSITION REPEATABILITY d_{rp} INDUCED BY THE MEASUREMENT NOISE OF THE GAUGE WITH $\gamma = 45^\circ$ AND 85°

	$\gamma = 45^\circ$		$\gamma = 85^\circ$	
	$\theta_{rp} (^\circ)$	$d_{rp} (\mu\text{m})$	$\theta_{rp} (^\circ)$	$d_{rp} (\mu\text{m})$
Mean	$\approx 10^{-4}$	48	$\approx 10^{-4}$	229
Max	$\approx 10^{-3}$	107	$\approx 10^{-3}$	431

Robustness was quantified using orientation repeatability (θ_{rp}) and position repeatability (d_{rp}), defined as the standard deviation of the orientation and position errors across the ten noisy trials for each pose. More details about the mathematical definitions can be found in Section III-E of the Supplementary Material.

The results were summarized in Table II, where the orientation repeatability of both gauges were in the order of 10^{-4} to 10^{-3} degrees, demonstrating the effectiveness of the denoising effect in the linear regression after RDP for segment direction vector estimation. The optimal gauge ($\gamma = 45^\circ$) exhibits high robustness to measurement noise. The mean position repeatability of $48 \mu\text{m}$ is within the intrinsic repeatability of the LPF, indicating that the method does not amplify sensor noise. By contrast, the suboptimal gauge ($\gamma = 85^\circ$) yielded significantly worse position repeatability, with a mean of $229 \mu\text{m}$, which is more than four times the noise level of the sensor. This clearly demonstrates that an improper gauge design can amplify measurement noise and degrade system performance. These simulation findings, which will be validated experimentally in Section V, underscore the critical importance of geometric optimization for achieving robust pose estimation and confirm that the proposed method maintains stability under expected sensor noise.

B. Robustness Against Structured Noise

To evaluate robustness against structured noise, two common scenarios encountered in measurement systems were considered: occlusion, which is frequent in cluttered workspaces, and specular reflection, which occurs on highly polished surfaces that lack the diffuse reflection necessary for accurate LPF measurements. Both phenomena result in segments of invalid data within the captured profile, potentially compromising the accuracy of the pose estimation. Thus, the robustness of the proposed method was validated by simulating profiles containing varying levels of invalid data points.

In these simulations, five distinct profiles were generated from five random LPF poses using the optimal gauge design ($\gamma = 45^\circ$). Each of these profiles was then subjected to seven levels of structured noise, where the percentage of invalid data points was varied from 0% to 60%. To accommodate the missing data, the proposed algorithm first performs an interpolation step on the profile before proceeding with the pose calculation. The resulting orientation errors ($\Delta\theta$) and position errors (Δd) were calculated for all successful estimation cases and are reported in Table III.

TABLE III

MEANS AND STANDARD DEVIATIONS (STD) OF THE ORIENTATION ERRORS $\Delta\theta$ AND POSITION ERRORS Δd OF THE POSE ESTIMATION WITH $\gamma = 45^\circ$ UNDER DIFFERENT PERCENTAGES OF INVALID DATA IN THE CAPTURED PROFILE

Invalid Data	$\Delta\theta (\text{deg})$		$\Delta d (\mu\text{m})$		Failure
	Mean	STD	Mean	STD	
0%	0.3	0.01	0.3	0.04	0/5
10%	0.7	0.7	0.6	0.6	0/5
20%	0.7	0.7	0.6	0.6	0/5
30%	322.7	158.7	204.6	94.2	2/5
40%	618.6	197.2	459.9	148.0	3/5
50%	1185.1	388.9	843.2	254.5	3/5
60%	-	-	-	-	5/5

Note: “-” represents no results due to complete failure.

The results show that for invalid data levels ranging from 0% to 20%, the pose estimation errors remained low and increased only marginally, indicating that sparse invalid data can be effectively compensated for by the interpolation routine. However, a significant performance degradation was observed at the 30% level, where two of the five trials failed, and the errors for the successful cases began to increase substantially. As the proportion of invalid data continued to increase, both the magnitude of the errors and failure rate escalated, culminating in a 100% failure rate (five out of five trials) at the 60% noise level.

Fig. 5 provides a visual explanation for this performance trend. As shown in Fig. 5(a), when only 10% of the profile data are invalid, the interpolation remains accurate, leading to correct corner detection and, consequently, to minimal estimation error. By contrast, Fig. 5(b) illustrates the effect of 40% invalid data, where poor interpolation, particularly in the vicinity of the corners, leads to inaccurate corner detection. This failure to correctly identify the key features of the profile ultimately causes the estimation algorithm to fail. On the basis of these results, the proposed method can be considered robust to structured noise levels of up to 20%, with performance degrading up to a 50% noise level before complete failure. In all successful estimation trials, the position errors remained at or below 1 mm, demonstrating the resilience and effectiveness of the method for 6-dof pose estimation under these challenging conditions.

C. Robustness in Cluttered Workspaces

In many robotic applications, the field of view of the LPF may include obstacles in the workspace. These extraneous objects can be unintentionally captured in the scan data, introducing significant errors into the pose estimation process. To mitigate this issue and ensure robust operation in cluttered environments, a gauge-identification algorithm was developed to automatically identify and filter out nongauge data points from the raw profile.

This algorithm functions by analyzing the geometric properties of the profile. It identifies the boundaries of the gauge by detecting discontinuities, which manifest as abrupt changes in the tangent slope of the profile. The contiguous segment of data between these detected boundary points is classified as the gauge region. All data points outside this region are considered

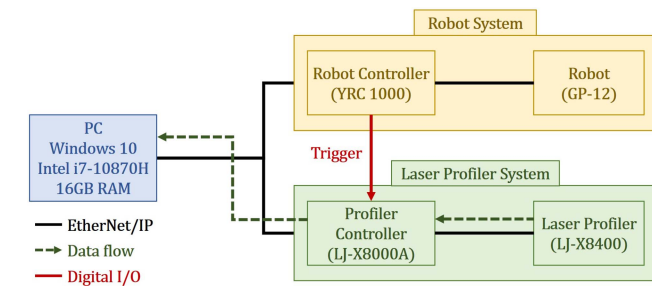


Fig. 7. Mechatronics communication of the proposed LPF-based system.

nongauge data, assumed to originate from obstacles, and are subsequently discarded from the calculation. The performance of this filtering algorithm was validated in a real-world experiment. Obstacles were deliberately placed in close proximity to the gauge, as shown in the experimental setup in Fig. 5(c). Fig. 5(d) illustrates the result of the gauge-identification process on the resulting scan. The algorithm successfully detects the border points of the gauge, correctly isolating the relevant gauge data (highlighted in red) from the obstacle-induced data (shown in blue). This successful segmentation confirms the effectiveness of the algorithm and demonstrates its capability to enable robust pose estimation even within cluttered workspaces.

V. EXPERIMENTAL VALIDATION

A. Experimental Setup

This section details the real-world experiments conducted to evaluate the performance of the proposed 6-dof pose estimation method. The experimental setup, as shown in Fig. 6(a), consists of the primary measurement system and validation instrumentation. The core of the system is a Keyence LJ-X8400 2D LPF, which has the repeatability of $5\ \mu\text{m}$ in the scan-line direction (x -axis) and $10\ \mu\text{m}$ in the depth direction (z -axis). The LPF was mounted on the flange of a Yaskawa GP-12, a six-axis industrial robot with a positioning repeatability of $0.02\ \text{mm}$. More detailed specifications can be found in Section IV-A of the Supplementary Material. Two custom gauges with different geometries were fabricated to validate both the proposed method and the conclusions of the sensitivity analysis from Section IV. To minimize the measurement noise caused by specular reflections from the metallic surfaces, the gauges were subjected to a blackening treatment to create a more diffuse finish.

Fig. 7 shows the mechatronics communication of the system. The LPF is connected to its controller via EtherNet/IP, while the controller's digital I/O is linked to the robot controller's digital I/O. The robot manipulator is connected to its controller, which then connects to a PC, where the pose calculation is performed, via EtherNet/IP. The PC utilized in this work has an Intel i7-10870H CPU, 16 GB RAM, and Windows 10 operating system.

The experimental validation was structured in three parts. First, an experiment was conducted to validate the sensitivity analysis by directly comparing the estimation performance of the two gauge geometries. Second, the repeatability of the proposed

TABLE IV
MEAN AND MAXIMUM VALUES OF—(a) ORIENTATION ERRORS ($\Delta\theta$) AND POSITION ERRORS (Δd), AND (b) ORIENTATION REPEATABILITY (θ_{rp}) AND POSITION REPEATABILITY (d_{rp})—OF GAUGE WITH $\gamma = 45^\circ$ AND 85°

		$\gamma = 45^\circ$		$\gamma = 85^\circ$	
		$\Delta\theta$ ($^\circ$)	Δd (mm)	$\Delta\theta$ ($^\circ$)	Δd (mm)
(a)	Mean	1.296	2.763	1.944	4.236
	Max.	1.447	4.426	3.046	8.446
		θ_{rp} ($^\circ$)	d_{rp} (mm)	θ_{rp} ($^\circ$)	d_{rp} (mm)
(b)	Mean	0.002	0.004	0.091	0.032
	Max.	0.003	0.005	0.117	0.071

6-dof pose estimation method was evaluated. Third, the practical utility of the method was demonstrated through its application to a hand-eye calibration task. For the sensitivity and repeatability experiments, a Vicon motion capture system comprising six cameras was used to provide ground-truth 6-dof pose data. This system provides a positioning accuracy of $0.017\ \text{mm}$ for its retroreflective markers. As shown in Fig. 6(a), markers were affixed to both the LPF and gauges to enable simultaneous tracking by the Vicon system.

B. Sensitivity Analysis Validation and Repeatability Evaluation

The experimental validation of the sensitivity analysis was guided by the simulation results reported in Section IV-A, which predicted that the gauge's susceptibility to fabrication errors is minimized when the geometric parameter γ is approximately 45° and is heightened as γ approaches 0° or 90° . Based on this finding, two gauges were fabricated with parameters $\{\phi, \gamma\}$ of $\{0^\circ, 45^\circ\}$ and $\{0^\circ, 85^\circ\}$, respectively. These specific geometries were chosen to allow for a direct performance comparison between a theoretically robust configuration and a more sensitive one. During the experiment, five distinct poses were measured for each gauge, and the resulting 6-dof pose estimates were compared against ground-truth measurements from the Vicon system.

The experimental outcomes, as summarized in Table IV(a), confirm the predictions of the simulation. The gauge with $\gamma = 45^\circ$ demonstrated superior accuracy, yielding an average orientation error of 1.296° and an average position error of $2.763\ \text{mm}$. By contrast, the gauge with $\gamma = 85^\circ$ resulted in substantially larger errors, with an average orientation error of 1.944° and an average position error of $4.236\ \text{mm}$. This trend was also reflected in the maximum observed errors, where the $\gamma = 85^\circ$ gauge exhibited a maximum position error nearly twice that of the $\gamma = 45^\circ$ gauge. These results empirically validate the expectation that the $\gamma = 45^\circ$ gauge outperforms the $\gamma = 85^\circ$ gauge, thus confirming the effectiveness of using sensitivity analysis to guide the geometric design.

In addition to the accuracy assessment, a repeatability experiment was conducted to validate the simulation findings from Section IV-A and to assess the robustness of the pose estimation method. The overall repeatability of the experimental system is influenced by both the measurement repeatability of the LPF

and the positioning repeatability of the robot manipulator. To isolate the impact of sensor measurement noise, the influence of the robot's motion was excluded by keeping the manipulator stationary during measurement sequences. For this test, the robot was moved to each of the five target poses. At each static pose, ten consecutive measurements were performed. The mean repeatability was then calculated across the five poses, with the results summarized in Table IV(b). With the optimal gauge ($\gamma = 45^\circ$), the mean repeatability was 0.002° for orientation and 0.004 mm for position. The results for the $\gamma = 85^\circ$ gauge were consistent with the simulation, showing less favorable mean repeatability values of 0.091° in orientation and 0.032 mm in position. However, the position repeatability for the less optimal gauge was still within the same order of magnitude as the intrinsic sensor repeatability of the LPF. This experiment further demonstrates the robustness of the proposed pose estimation method and underscores the importance of proper gauge design in achieving high-performance results.

C. Using the Proposed System for Hand–Eye Calibration

A primary application for high-accuracy, real-time pose estimation is hand–eye calibration. This process is critical for robotic systems but presents challenges for 2-D LPFs. Traditional hand–eye calibration methods solve the $\mathbf{AX} = \mathbf{XB}$ equation [37], which requires at least two distinct robot poses. In this formulation, $\mathbf{A}$ represents the transformation between two robot flange poses ($\mathbf{A} = \mathbf{F}_1^{-1}\mathbf{F}_2$), and $\mathbf{B}$ is the transformation between the corresponding sensor poses relative to a fixed calibration target ($\mathbf{B} = \mathbf{V}_1\mathbf{V}_2^{-1}$). The unknown to be solved is $\mathbf{X}$, the rigid transformation between the robot's flange and the sensor frame.

While robot flange poses ($\mathbf{F}_1$ and $\mathbf{F}_2$) are readily available from the robot controller, determining the sensor poses ($\mathbf{V}_1$ and $\mathbf{V}_2$) is the main challenge for an LPF. Unlike a camera, which can compute its full 6-dof pose from a single image of a chessboard [28], [29], the 2-D profile of an LPF does not inherently provide enough information. The proposed method overcomes this fundamental limitation by enabling the direct estimation of the full 6-dof sensor pose, $\mathbf{V}$, from a single LPF scan of the custom gauge.

To evaluate the hand–eye calibration performance of the proposed system, the calibration results using Vicon along with a CMM were set up as the ground truth. The CMM used was Mitutoyo STRATO-Apex 574, which has a maximum permissible error of length measurement of $0.7 + 0.25L/100$ μm according to ISO 10360-2:2009. The pose measurement by Vicon with CMM is described as follows. First, Vicon measures the retroreflective markers mounted on the gauge and LPF, followed by constructing the frames defined by the markers, which are ${}^{\text{Vicon}}G_{\text{mark}}\mathbf{T}$ and ${}^{\text{Vicon}}E_{\text{mark}}\mathbf{T}$, respectively. Next, the transformation matrices ${}^G G_{\text{mark}}\mathbf{T}$ and ${}^E E_{\text{mark}}\mathbf{T}$ were established by measuring the markers using CMM. Thus, the pose of the LPF relative to the gauge frame ${}^G E\mathbf{T}$ can be computed as follows:

$${}^G E\mathbf{T} = {}^G G_{\text{mark}}\mathbf{T} {}^{\text{Vicon}}G_{\text{mark}}\mathbf{T}^{-1} {}^{\text{Vicon}}E_{\text{mark}}\mathbf{T} {}^E E_{\text{mark}}\mathbf{T}^{-1}. \quad (25)$$

Here, the CMM measured the LPF frame $\{\mathbf{E}\}$ by touching the origin with its tooltip and moving along the x - and z -axes

TABLE V
HAND–EYE CALIBRATION ERRORS USING VICON AND LPF WITH TWO CALIBRATION ALGORITHMS

	$\Delta\theta_{he} (^\circ)$	$\Delta d_{he} (\text{mm})$			
		$\Delta d_{he,x}$	$\Delta d_{he,y}$	$\Delta d_{he,z}$	$ \Delta d_{he} $
Tsai and Lenz's [28]	0.123	0.184	0.200	0.152	0.311
Zhuang et al.'s [29]	0.158	0.212	0.183	0.146	0.315
Mean	0.141	0.198	0.192	0.149	0.313

of the LPF frame, where the instant position of the tooltip in the LPF frame can be monitored using the real-time monitoring software of the LPF.

In the experiment, the optimal gauge ($\gamma = 45^\circ$) was used. The robot moved the LPF to nine different poses. At each pose, the 6-dof position of the sensor was estimated using the proposed method. The nine poses were then input into two widely used calibration algorithms: Tsai and Lenz [28] and Zhuang et al.'s [29] methods. The resulting hand–eye transformations were compared with the Vicon/CMM ground truth. As shown in Table V, the results demonstrate a high degree of accuracy. The average orientation error ($\Delta\theta_{he}$) across both algorithms was 0.141° , and the average position error ($|\Delta d_{he}|$) was 0.313 mm. These low error values confirm that the proposed LPF-based pose estimation method is sufficiently accurate and robust for direct application in standard hand–eye calibration procedures, effectively enabling LPFs to be used as seamlessly as cameras for this critical task.

VI. BENCHMARK FOR THE CAMERA-BASED METHOD

To contextualize the performance of the proposed LPF-based system, a series of benchmark experiments was conducted to compare it against a traditional camera-based approach. The comparison focused on three key performance areas: 1) 6-dof pose estimation accuracy, 2) computational efficiency, and 3) hand–eye calibration performance. The camera-based system consisted of a Basler acA2500-14gc camera (2590×1942 resolution) and a standard 8×11 chessboard with squares of 20 mm^2 size.

A direct comparison of the estimated pose matrices ${}^G E\mathbf{T}$ obtained from (25) was not feasible. Whereas the Vicon system could provide a ground-truth pose for the LPF, establishing an equivalent ground truth for the camera is problematic. Specifically, using a CMM to measure the transformation between the camera's optical center (the origin of its frame) and the Vicon markers is impractical, as one cannot visually confirm when the CMM probe has reached the nonphysical origin point within the camera. To ensure a fair and meaningful comparison, we instead used the reconstruction error as the primary accuracy metric [38]. This metric evaluates how well the estimated pose of each system can reproject key feature points back to their known ground-truth locations in the world frame. For the camera, these features are the corners of the chessboard. For the LPF, the features are the corner points of the gauge profile. The reconstruction error for the k th corner point of the LPF, e_k , is defined as the shortest distance from the reconstructed corner

TABLE VI

BENCHMARK EVALUATION OF THE CAMERA-BASED METHOD BY COMPARING THE MEANS AND STANDARD DEVIATIONS (STD) OF THE RECONSTRUCTION ERRORS AND TIME COST PER POSE OF TEN POSE ESTIMATIONS

	RE (mm)		Time Cost (sec)	
	Mean	STD	Mean	STD
Camera	0.437	0.446	0.641	0.067
LPF	0.241	0.056	0.056	0.004
Improvement	44.9%	87.4%	91.3%	94.0%

TABLE VII

BENCHMARK EVALUATION OF THE CAMERA-BASED METHOD BY COMPARING THE MEANS AND STANDARD DEVIATIONS (STD) OF THE RESIDUAL ERRORS OF HAND-EYE CALIBRATION OF TEN POSE ESTIMATIONS

	Tsai and Lenz's [28]		Zhuang et al.'s [29]	
	Mean	STD	Mean	STD
Camera	1.753	0.872	1.713	0.913
LPF	0.446	0.213	0.449	0.180
Improvement	74.6%	75.6%	73.8%	80.3%

point, ${}^G\mathbf{p}_k$, to its corresponding true intersection line, $\mathbf{L}_k$, which lies on the profile plane $\mathbf{S}_{\text{LPF}}$

$$e = \frac{1}{N} \sum_{k=1}^N e_k = \frac{1}{N} \sum_{k=1}^N \| {}^G\mathbf{p}_k - {}^G(\mathbf{L}_k \cap \mathbf{S}_{\text{LPF}}) \|_2 \quad (26)$$

where $\mathbf{S}_{\text{LPF}}$ is the xz -plane of the estimated pose ${}^G\mathbf{T}$. ${}^G\mathbf{p}_k$ is the reconstructed corner point obtained by ${}^G\mathbf{p}_k = {}^G\mathbf{T}^E\mathbf{p}_k$, which transforms the corner point in the captured profile (${}^E\mathbf{p}_k$), to the gauge frame with the estimated pose (${}^G\mathbf{T}$).

The reconstruction errors for both systems are summarized in Table VI. While the camera-based method delivered performance sufficient for many general-purpose applications, the proposed LPF-based method demonstrated markedly superior accuracy. The LPF system achieved a mean reconstruction error of 0.241 mm with a standard deviation of 0.056 mm, representing a 44.9% reduction in mean error and an 87.4% reduction in standard deviation compared with the camera system. This result highlights the superior accuracy and robustness of the LPF-based method, making it particularly well suited for high-precision robotic tasks.

The computational efficiencies of the two methods were also compared. As shown in Table VI, the proposed LPF-based method provides a dramatic speed advantage. It achieved 0.056 s per pose with only 0.004 s standard deviations, indicating 91.3% and 94.0% reductions, respectively, compared with the camera-based method. This efficiency and stability are attributed to the lightweight nature of the LPF data and the insensitivity of the LPF to lighting variations, respectively. The low time cost and minimal variance confirm the suitability of the method for high-speed, real-time applications.

Finally, the hand-eye calibration performance of both methods was benchmarked. As a Vicon-based ground truth was not available for the camera, performance was evaluated using the residual error of the $\mathbf{AX} = \mathbf{XB}$ solution, where lower values

indicate a more consistent calibration. The residual error, E , was calculated using the Frobenius norm

$$E = \|(\mathbf{AX})^{-1}\mathbf{XB}\|_F. \quad (27a)$$

The results, presented in Table VII, show that the LPF-based method achieved a significant improvement, reducing the mean residual error by approximately 70% to 80% for both Tsai and Lenz's [28] and Zhuang et al.'s [29] algorithms. This demonstrates the superior stability and practical effectiveness of the proposed method for high-precision hand-eye calibration.

VII. CONCLUSION

This article introduced a novel, real-time 6-dof pose estimation method for a 2-D LPF that uses a single scan of a custom-designed four-plane gauge. By sequentially solving for the 3-dof orientation and position from the captured profile, the method provides a high-accuracy, low-latency solution suitable for a wide range of robotic applications. A key contribution of this work is the sensitivity analysis that guided the geometric design of the gauge, identifying that a symmetric configuration ($\phi = 0^\circ$) with an intermediate plane divergence ($\gamma \approx 45^\circ$) minimizes sensitivity to fabrication errors and sensor noise.

The robustness of the method was extensively validated through simulation and experiment. The method was shown to be resilient to unstructured noise, with the optimized gauge design preventing the amplification of intrinsic sensor noise. It also demonstrated tolerance to structured noise, successfully operating with up to 20% invalid data from simulated occlusions or reflections. A practical gauge-identification algorithm was also developed and proven to effectively filter out environmental clutter.

Experimental validation against a Vicon motion capture system confirmed the predictions of the sensitivity analysis. The optimized gauge ($\gamma = 45^\circ$) consistently outperformed the sub-optimal design ($\gamma = 85^\circ$) with a mean error of 1.296° in orientation and 2.763 mm in position and a mean repeatability of only 0.002° and 0.004 mm. These results validate both the accuracy of the proposed method and the effectiveness of the sensitivity-guided gauge design. The practical utility of the method was further demonstrated in a hand-eye calibration task, where it achieved an orientation error of 0.141° and position error of 0.313 mm with a high-precision CMM/Vicon ground truth, confirming its seamless compatibility with standard calibration routines.

Finally, the comprehensive benchmark against the conventional camera-based system highlights the significant advantages of the proposed method. The LPF system was not only more accurate, reducing the reconstruction error by more than 44% with 0.241 mm, but also an order of magnitude faster, with a 91.3% reduction in computation time to only 56 ms per pose. Furthermore, it improved hand-eye calibration stability, reducing residual errors by more than 70%. The proposed method achieved better pose accuracy, computational efficiency, and calibration stability, establishing the proposed LPF-based approach as a superior alternative for applications demanding high speed and precision.

Future work will focus on applying this high-performance pose estimation system to real-time robotic control tasks, such as trajectory tracking and dynamic compensation, where its high sampling rate and computational efficiency can be fully leveraged.

ACKNOWLEDGMENT

The authors would like to thank the National Chung-Shan Institute of Science and Technology (NCSIST), Taiwan, for the various useful consults and support for this work.

REFERENCES

- [1] X. Niu, Y. Peng, Y. Dai, Q. Chen, C. Guo, and Q. Zhang, "Camera-based lane-aided multi-information integration for land vehicle navigation," *IEEE/ASME Trans. Mechatron.*, vol. 28, no. 1, pp. 152–163, Feb. 2023.
- [2] M. W. M. G. Dissanayake, P. Newman, S. Clark, H. Durrant-Whyte, and M. Csorba, "A solution to the simultaneous localization and map building (SLAM) problem," *IEEE Trans. Robot. Autom.*, vol. 17, no. 3, pp. 229–241, Jun. 2001.
- [3] C.-H. Chien, C.-C. J. Hsu, W.-Y. Wang, and H.-H. Chiang, "Indirect visual simultaneous localization and mapping based on linear models," *IEEE Sensors J.*, vol. 20, no. 5, pp. 2738–2747, Mar. 2020.
- [4] Z. Zhou, L. Wang, and V. Popescu, "A partially-sorted concentric layout for efficient label localization in augmented reality," *IEEE Trans. Vis. Comput. Graphics.*, vol. 27, no. 11, pp. 4087–4096, Nov. 2021.
- [5] H. Chen, D. Sun, J. Yang, and J. Chen, "Localization for multirobot formations in indoor environment," *IEEE/ASME Trans. Mechatron.*, vol. 15, no. 4, pp. 561–574, Aug. 2010.
- [6] F. Nie, W. Zhang, Y. Wang, Y. Shi, and Q. Huang, "A forest 3-D lidar SLAM system for rubber-tapping robot based on trunk center atlas," *IEEE/ASME Trans. Mechatron.*, vol. 27, no. 5, pp. 2623–2633, Oct. 2022.
- [7] Y. Wu and J. Zhao, "A robust and precise LiDAR-inertial-GPS odometry and mapping method for large-scale environment," *IEEE/ASME Trans. Mechatron.*, vol. 27, no. 6, pp. 5027–5036, Dec. 2022.
- [8] A. Kendall, M. Grimes, and R. Cipolla, "PoseNet: A convolutional network for real-time 6-DOF camera relocalization," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2015, pp. 2938–2946.
- [9] J. Valentin, M. Nießner, J. Shotton, A. Fitzgibbon, S. Izadi, and P. Torr, "Exploiting uncertainty in regression forests for accurate camera relocalization," in *Proc. 2015 IEEE Conf. Comput. Vis. Pattern Recognit.*, 2015, pp. 4400–4408.
- [10] L. Liu, H. Li, and Y. Dai, "Efficient global 2D-3D matching for camera localization in a large-scale 3D map," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 2372–2381.
- [11] H. Lim, S. N. Sinha, M. F. Cohen, and M. Uyttendaele, "Real-time image-based 6-DOF localization in large-scale environments," in *Proc. 2012 IEEE Conf. Comput. Vis. Pattern Recognit.*, 2012, pp. 1043–1050.
- [12] B. Bescos, J. M. Fàcil, J. Civera, and J. Neira, "DynaSLAM: Tracking, mapping, and inpainting in dynamic scenes," *IEEE Robot. Autom. Lett.*, vol. 3, no. 4, pp. 4076–4083, Oct. 2018.
- [13] J. Shotton, B. Glocker, C. Zach, S. Izadi, A. Criminisi, and A. Fitzgibbon, "Scene coordinate regression forests for camera relocalization in RGB-D images," in *Proc. 2013 IEEE Conf. Comput. Vis. Pattern Recognit.*, 2013, pp. 2930–2937.
- [14] C. Yao, S. He, H. Chen, X. Zhang, and Z. Wang, "Pose estimation of nonoverlapping FoV cameras for shield tunnel convergence measurement," *Measurement*, vol. 242, 2025, Art. no. 116101.
- [15] Z. Liu, B. Guan, Y. Shang, Y. Bian, P. Sun, and Q. Yu, "Stereo event-based, 6-DoF pose tracking for uncooperative spacecraft," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5607513.
- [16] J. Jiao et al., "LiteVLoc: Map-lite visual localization for image goal navigation," in *Proc. 2025 IEEE Int. Conf. Robot. Autom.*, 2025, pp. 5244–5251.
- [17] H. Xu et al., "SurgRIPE challenge: Benchmark of surgical robot instrument pose estimation," *Med. Image Anal.*, vol. 105, 2025, Art. no. 103674.
- [18] M. Liu, J. Guo, L. Deng, S. Wang, and H. Wang, "Enhanced vision-based 6-DoF pose estimation for robotic rebar tying," *Automat. Construction*, vol. 171, 2025, Art. no. 105999.
- [19] R. Li, N. Ding, Y. Zhao, and H. Liu, "Real-time trajectory position error compensation technology of industrial robot," *Measurement*, vol. 208, 2023, Art. no. 112418.
- [20] C. Moeller et al., "Real time pose control of an industrial robotic system for machining of large scale components in aerospace industry using laser tracker system," *SAE Int. J. Aerosp.*, vol. 10, no. 2, pp. 100–108, 2017.
- [21] T. Shu, S. Gharaaty, W. Xie, A. Joubair, and I. A. Bonev, "Dynamic path tracking of industrial robots with high accuracy using photogrammetry sensor," *IEEE/ASME Trans. Mechatron.*, vol. 23, no. 3, pp. 1159–1170, Jun. 2018.
- [22] E. Zakeri and W.-F. Xie, "Robust photogrammetry sensor-based real-time pose control of industrial robots," *IEEE/ASME Trans. Mechatron.*, vol. 29, no. 5, pp. 3509–3521, Oct. 2024.
- [23] R. A. Boby and S. K. Saha, "Single image based camera calibration and pose estimation of the end-effector of a robot," in *Proc. 2016 IEEE Int. Conf. Robot. Autom.*, 2016, pp. 2435–2440.
- [24] TR ISO, "13309: Manipulating industrial robots—informative guide on test equipment and metrology methods of operation for robot performance evaluation in accordance with ISO 9283," International Organization for Standardization, Geneva, Switzerland, 1995.
- [25] J. M. S. Motta, G. C. De Carvalho, and R. McMaster, "Robot calibration using a 3D vision-based measurement system with a single camera," *Robot. Comput.- Integr. Manuf.*, vol. 17, no. 6, pp. 487–497, 2001.
- [26] V. H. Pinto, A. Amorim, L. Rocha, and A. P. Moreira, "Enhanced performance real-time industrial robot programming by demonstration using stereoscopic vision and an IMU sensor," in *Proc. 2020 IEEE Int. Conf. Auton. Robot. Syst. Competitions*, 2020, pp. 108–113.
- [27] V. Lertpiriyasuwat and M. C. Berg, "Adaptive real-time estimation of end-effector position and orientation using precise measurements of end-effector position," *IEEE/ASME Trans. Mechatron.*, vol. 11, no. 3, pp. 304–319, Jun. 2006.
- [28] R. Y. Tsai and R. K. Lenz, "A new technique for fully autonomous and efficient 3D robotics hand/eye calibration," *IEEE Trans. Robot. Autom.*, vol. 5, no. 3, pp. 345–358, Jun. 1989.
- [29] H. Zhuang, Z. S. Roth, and R. Sudhakar, "Simultaneous robot/world and tool/flange calibration by solving homogeneous transformation equations of the form $AX=YB$," *IEEE Trans. Robot. Autom.*, vol. 10, no. 4, pp. 549–554, Aug. 1994.
- [30] Q. Zhang and R. Pless, "Extrinsic calibration of a camera and laser range finder (improves camera calibration)," in *Proc. 2004 IEEE/RSJ Int. Conf. Intell. Robots Syst. (IEEE Cat. No. 04CH37566)*, 2004, pp. 2301–2306.
- [31] R. Unnikrishnan and M. Hebert, "Fast extrinsic calibration of a laser rangefinder to a camera," Robotics Inst., Carnegie Mellon Univ., Pittsburgh, PA, USA, Tech. Rep. CMU-RI-TR-05-09, 2005.
- [32] F. B. Carlson, R. Johansson, and A. Robertsson, "Six DoF eye-to-hand calibration from 2D measurements using planar constraints," in *Proc. 2015 IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2015, pp. 3628–3632.
- [33] S. Sharifzadeh, I. Biro, and P. Kinnell, "Robot hand-eye calibration of 2D laser sensors using a single-plane calibration artefact," *Robot. Comput.- Integr. Manuf.*, vol. 61, 2020, Art. no. 101823.
- [34] T. S. Lembono, F. Suárez-Ruiz, and Q.-C. Pham, "Scalar: Simultaneous calibration of 2-D laser and robot kinematic parameters using planarity and distance constraints," *IEEE Trans. Autom. Sci. Eng.*, vol. 16, no. 4, pp. 1971–1979, Oct. 2019.
- [35] J. Xu, J. L. Hoo, S. Dritsas, and J. G. Fernandez, "Hand-eye calibration for 2D laser profile scanners using straight edges of common objects," *Robot. Comput.- Integr. Manuf.*, vol. 73, 2022, Art. no. 102221.
- [36] D. H. Douglas and T. K. Peucker, "Algorithms for the reduction of the number of points required to represent a digitized line or its caricature," *Cartographica: Int. J. Geographic Inf. Geovisualization*, vol. 10, no. 2, pp. 112–122, 1973.
- [37] B. N. Datta, *Numerical Methods for Linear Control Systems: Design and Analysis*. Boston, MA, USA: Elsevier, 2004.
- [38] Hartley, "An algorithm for self calibration from several views," in *Proc. 1994 Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 1994, pp. 908–912.



Tao Chen (Member, IEEE) is currently working toward the Ph.D. degree in mechanical engineering at National Taiwan University, Taipei, Taiwan.

His research interests include sensing systems and manipulation.



Jia-Xin Liu received the M.S. degree in mechanical engineering from National Taiwan University, Taipei, Taiwan, in 2023.

His research interests include manipulation and kinematic calibration.



Pei-Chun Lin (Senior Member, IEEE) received the Ph.D. degree in mechanical engineering from the University of Michigan, Ann Arbor, MI, USA, in 2005.

He is currently a distinguished professor in the Department of Mechanical Engineering, National Taiwan University, Taipei, Taiwan. His research interests include legged locomotion, mechanism design, sensing systems, and manipulation.



Yao-Yang Tsai received the Ph.D. degree in engineering from the University of Tokyo, Tokyo, Japan.

He is currently an Associate Professor with the Department of Mechanical Engineering, National Taiwan University. His research interests include micromachining, nontraditional machining processes, and intelligent manufacturing and its applications.